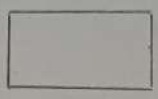
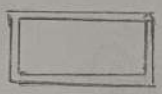
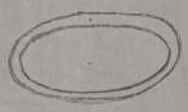
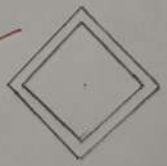
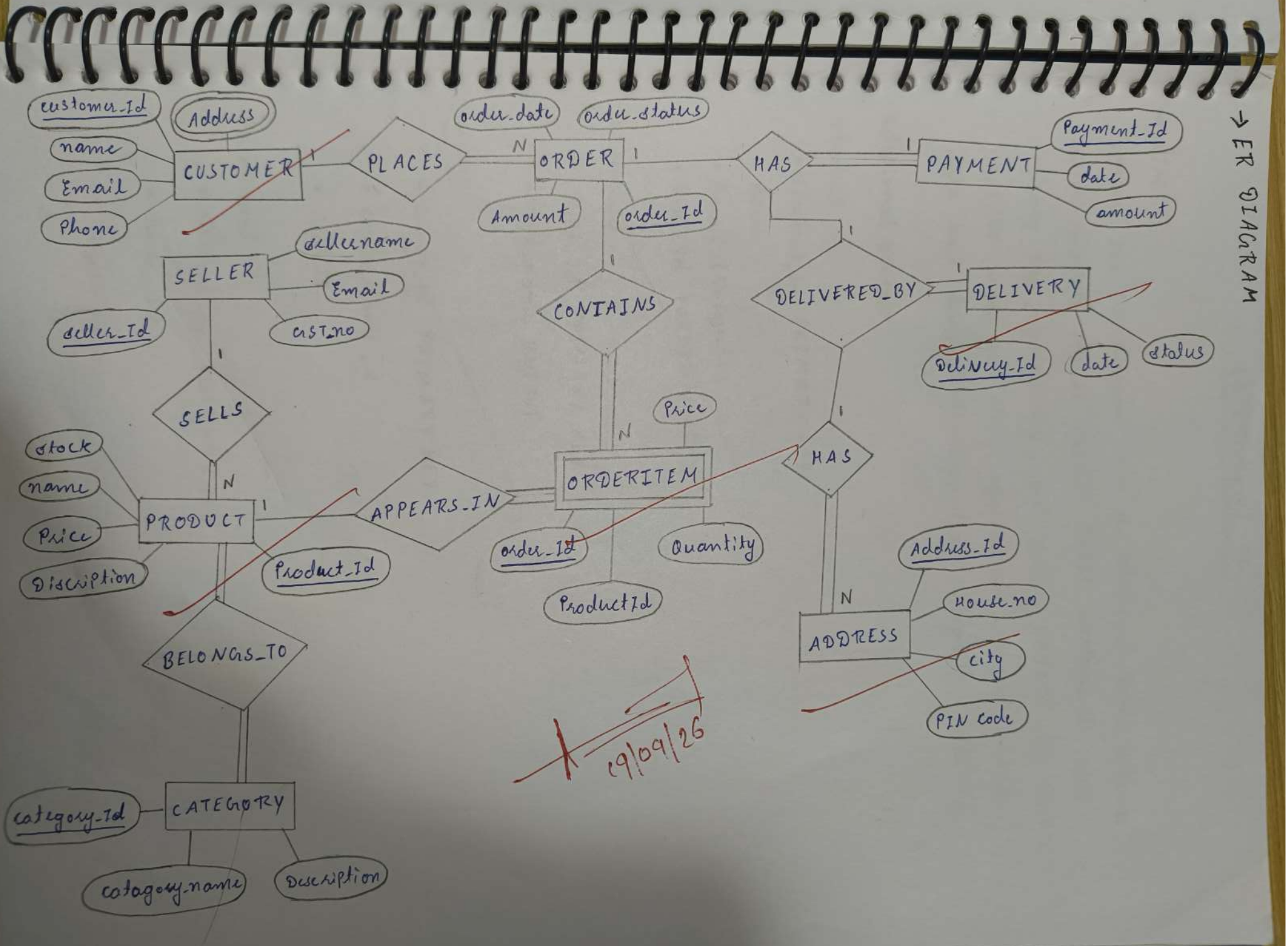
EXPERIMENT - 01

→ AIM

Design an ER diagram for an indian e-commerce platform with entities Customer, Product, OrderItem, Seller, Category, Payment, Delivery, Address. Specify Primary keys, composite and multi-valued attributes, Weak entities, Specialization of Product, and participation constraints.

- Representation of ER diagram components

1. Strong Entity - 
2. Weak Entity - 
3. Attribute - 
4. Multi-valued attribute - 
5. composite Attribute - 
6. Derived Attribute - 
7. Relationship - 
8. Identifying Relationship - 
9. Partial Participation -
10. Total Participation -
11. Cardinality - 1, N, M


→ ER DIAGRAM
customer_Id
Address
name
Email
Phone
CUSTOMER
PLACES
order_date
order_status
N
ORDER
1
Amount
order_Id
HAS
Payment_Id
PAYMENT
date
amount
sellername
SELLER
Email
seller_Id
cust_no
CONTAINS
DELIVERED_BY
DELIVERY
DeliVery_Id
date
status
SELLS
Price
N
stock
name
Price
Discription
PRODUCT
APPEARS_IN
ORDERITEM
HAS
order_Id
Quantity
Product_Id
ProductId
Address_Id
House_no
N
ADDRESS
city
PIN code
BELONGS_TO
19/09/26
category_Id
CATEGORY
catagory_name
Description

EXPERIMENT-02

→ AIM

Convert the ER diagram into a relational table schema. Write complete MYSQL CREATE TABLE statements with Primary key, foreign key, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

Relational schema

Student

- Student_id PRIMARY KEY
- name
- Email (Unique)
- Dept_id FOREIGN KEY

Department

- Dept_id PRIMARY KEY
- Dept_name UNIQUE

Course

- Course_id PRIMARY KEY
- course_name, €
- credits

Enrollment

- student_id PRIMARY KEY
- course_id
- Enroll_date

Source code (MY SQL)

```
CREATE TABLE Department (
        Dept_id INT PRIMARY KEY,
        Dept_name VARCHAR(50) NOT NULL UNIQUE
);

CREATE TABLE students (
        student_id INT PRIMARY KEY,
        name VARCHAR(50) NOT NULL,
        Email VARCHAR(100) UNIQUE,
        Dept_id INT
        FOREIGN KEY (Dept_id)
            REFERENCES Department (Dept_id)
            ON DELETE SET NULL
);

CREATE TABLE course(
        course_id INT PRIMARY KEY,
        course_name VARCHAR(100) NOT NULL,
        credits INT NOT NULL
);
```

```
CREATE TABLE Enrollment (
        student_id INT,
        course_id INT,
        Enroll_date DATE NOT NULL,
        PRIMARY KEY (student_id, course_id),

        FOREIGN KEY (student_id)
            REFERENCES student (student_id)
            ON DELETE CASCADE,

        FOREIGN KEY (course_id)
            REFERENCES course (course_id)
            ON DELETE CASCADE
);
```

Insert Sample Data

```
INSERT INTO Department VALUES
(1, 'CSE'),
(2, 'ECE');

INSERT INTO students VALUES
(101, 'Rohit', 'rohit@gmail.com', 1),
(102, 'Rohan', 'rohan@gmail.com', 2);

INSERT INTO course VALUES
(201, 'DBMS', 4),
(202, 'Python', 3);

INSERT INTO Enrollment VALUES
(101, 201, '2026-08-19'), (101, 202, '2026-08-19'),
(102, 201, '2026-07-20');
```

→ OUTPUT

SELECT * FROM students;

student_id	name	Email	Dept_id
1	Rohit	rohit@gmail.com	1
2	Rohan	rohan@gmail.com	2

SELECT * FROM Enrollment;

student_id	course_id	Enroll_date
101	201	2026-08-19
101	202	2026-08-19
102	201	26-07-20

SELECT * FROM Department.

Dept_id	Dept_name
1	CSE
2	ECE

19/09/26.